



INDIA RUNS × REDROB AI · TRACK 01

ATLAS

Intelligent Candidate Discovery & Ranking

The Data & AI Challenge · Team **IndiaRuns**

Retrieval finds the candidates. An LLM judges the few that matter. A monotone-constrained blender fuses the evidence. A structural gate guarantees no trap survives — all frozen offline so the graded step is a 73-second pure-numpy pass.

100,000 real candidates

73.5 s · CPU-only · offline

0 honeypots in top-100

Validator: **valid**

Recruiters drown in profiles — and still **miss the right person.**

- ◆ Keyword filters match **words, not fit**. They rank by skill-count, so a buzzword-stuffed profile beats a quiet specialist who actually shipped the system.
- ◆ The cost is two-sided: **great candidates stay invisible**, and recruiters burn hours on confident-looking non-fits.
- ◆ Redrob's bet is an **AI brain for hiring** — read a job description and rank like a great recruiter would, by understanding **who fits**, not who matched the most strings.

And this dataset makes that hard **on purpose.**

The dataset is a **trap** — it punishes keyword-counting by design.

20.4%

have an AI/ML **skill keyword** in their profile

1.0%

have an AI/ML **job title** — a 19.4-pt gap

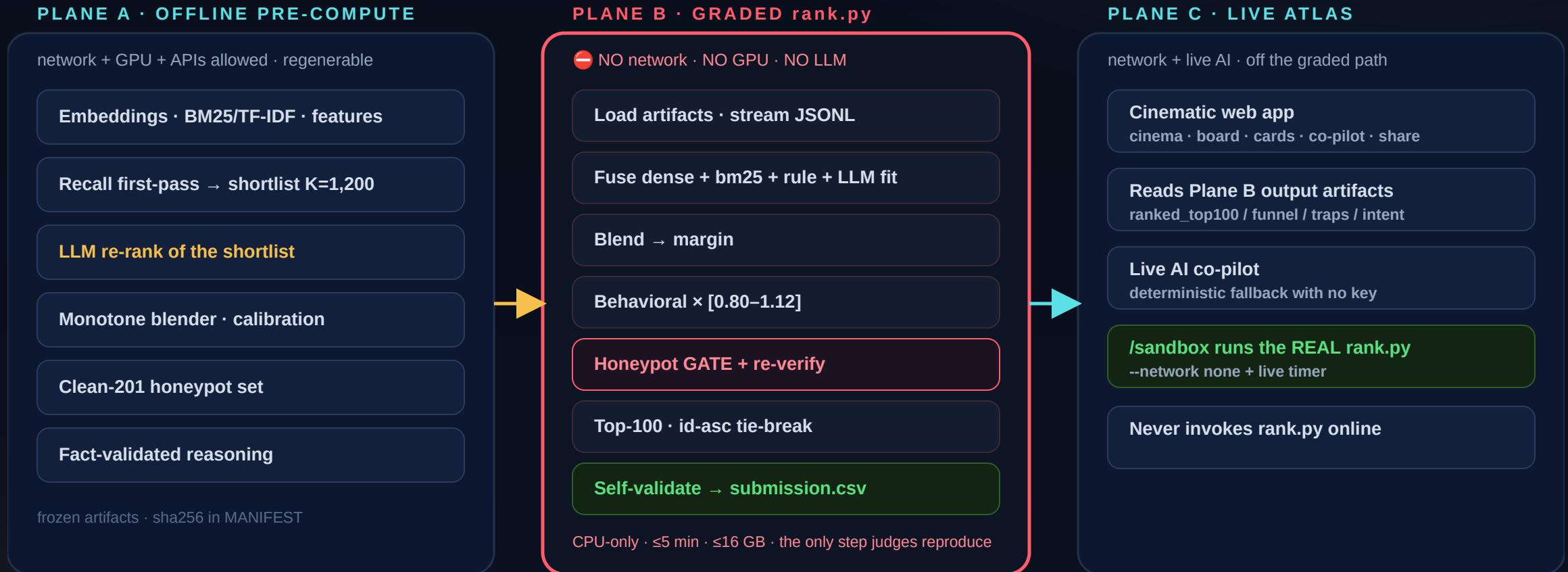
#1

the baseline ranks an **HR Manager** first — purely on skill count

- ◆ That gap is overwhelmingly **non-AI roles that stuffed AI keywords** into their skills array.
- ◆ Planted **honeypots** — impossible-on-the-numbers profiles — sit in the pool. More than **10 in your top-100 is an automatic disqualification**.

The challenge: **beat the keyword instinct**, and **don't step on a landmine**.

Three planes, one inviolable boundary.



The one rule: no network, GPU, or LLM call ever sits on the graded path — the single most defensible point of the whole submission.

The hybrid funnel: **recall** → **judge** → **fuse** → **gate**.

- ◆ **1 · Recall first-pass.** $0.30 \cdot \text{dense} + 0.25 \cdot \text{bm25} + 0.45 \cdot \text{rule}$ over all 100K → top **1,200**, force-include AI-titled & strong-evidence, drop honeypots.
- ◆ **2 · LLM re-rank** the shortlist only — where **~80% of the metric lives**. 299 real judgments on the top-300.
- ◆ **3 · Blend / LTR fusion** — monotone-constrained. We **ship the safer one**.
- ◆ **4 · Behavioral** × [0.80–1.12] — modulates, never dominates. Sentinels never penalize.
- ◆ **5 · Honeypot hard-gate** + live re-verify of survivors.

Recall gate — measured, not assumed

100% of proxy-Tier-5 (**700**) and 100% of proxy-Tier-4 (**216**) land inside the 1,198-id shortlist. Nothing good is lost before the LLM sees it.

Ship the safer model

Fixed-weight blend won 5-fold CV-NDCG@10 **1.0000** vs learned LTR **0.766** → the blend ships. The monotone LTR path is fully built and reproducible.

```
# shortlisted (the only rows that reach top-100) base_fit = 0.50·S_llm + 0.20·S_dense + 0.18·S_rule + 0.12·S_bm25 margin =
blend(features) → × anti_trap_caps → × behavioral → honeypot GATE
```

We rank by what you **did** — not what you **listed**.

- ◆ **Embed the career narrative**, weighting descriptions **above** the skills array — so stuffing the skills list cannot move the vector.
- ◆ **Evidence features read from descriptions**: shipped a ranking/search/recsys system, built embeddings retrieval, designed NDCG/MRR/MAP eval, deployed to production.

- **The role/skill-mismatch cap — × 0.15**

A non-engineering current title that lists an AI skill is multiplied down by **0.15**. This is the direct counter to the baseline's HR-Manager-#1 failure — and it acts as a **deterministic floor under the LLM**, so model optimism can never out-vote a structural disqualifier.

A Marketing Manager's campaign text embeds **far** from the role — **regardless of stuffed skills**.

Three deterministic layers — **no trap survives.**

- **Clean-201 hard exclude**

Rebuilt from high-precision structural impossibilities **only** — gated to `-inf`:

too_many_experts	167	career_sum_exceeds_yoe	24
expert_zero_duration	21	tenure_exceeds_company_age	3
career_span_exceeds_yoe	3	= exactly	201

- **Live re-verify + top-10 paranoia**

Every survivor reaching the top-300 is re-checked; each top-10 slot must show real shipping evidence.

- **The critical non-check**

Salary `min > max` fires on **18.9% of the pool** — it is a **dataset norm, not a trap**. Flagging it would delete a fifth of the candidates, including genuine fits. We deliberately do **not** flag it.

0

honeypots in top-100

0

honeypots in top-10

Receipts, not promises.

METRIC	RESULT	CAP
Wall-clock (full 100K)	73.5 s	300 s
Peak memory	2.01 GB	16 GB
Network / GPU at rank time	none	—
Validator	valid	—
Determinism	byte-identical	694f86dd...
Honeypots top-100 / top-10	0 / 0	≤10
Lift vs naive baseline	~1.00 / 0.07	NDCG@10

Tier distribution / 100K — T0 43,961 · T1 25,035 · T2 5,978 · T3 24,110 · T4 216 · T5 700

Top-10 — genuine shippers

- 1 Sr ML Eng @ Genpact AI · LinkedIn RAG ranking, 50M q/mo
 - 2 Search Eng @ PharmEasy · L2R search
 - 3 Recsys Eng @ CRED · LTR + RAG eval
 - 4 ML Eng @ Flipkart · prod RAG, 10M recsys
 - 5 Recsys Eng @ Zomato
 - 6 Search Eng @ Saarthi.ai · L2R @ Dream11, FAISS
 - 7 AI Eng @ Ola
 - 8 Sr DS @ Amazon
 - 9 Search Eng @ Meta
 - 10 Search Eng @ Haptik
- Real product-company ranking & search engineers — not keyword phantoms.

ATLAS — the ranking, made legible and trustworthy.

- ◆ **Ranking cinema** — a real-data particle funnel: 100,000 → 1,198 → the **201 traps combust** → crystallize to top-100 then top-10. Every burning dot is a real flagged id.
- ◆ **Evidence cards** — the verbatim CSV reasoning, candidate's own words highlighted and tagged to the matched requirement, with one honest concern.
- ◆ **Honeypot reveal drawer** — "caught before they could be hired," with salary-inversion shown as **not** a trap.
- ◆ **Recruiter co-pilot** — filter, compare, draft outreach. Honest status pill.
- ◆ **Shareable shortlist** — auto OG image + one-page top-10 PDF.

ATLAS · Top 100 · 100,000 screened · 201 traps removed · 73 s offline

Live AI

1

Sr ML Engineer · Genpact AI

product

98

Linkedin RAG ranking · 50M q/mo · resp 0.81 · open to work

2

Search Engineer · PharmEasy

product

96

L2R search · NDCG/MRR eval · ≤30d notice

3

Recsys Engineer · CRED

product

95

LTR + RAG eval · FAISS · verified email

×

HR Manager · 7 "expert" AI skills

honeypot: too_many_experts · excluded

0

co-pilot > "show product-company candidates open to work in ≤30 days"

Heavy AI everywhere it helps — **never** on the graded path.

● NVIDIA — Plane A

Hosted embeddings + a 70B-class instruct re-rank of the shortlist and fact-validated reasoning. Full path built; degrades cleanly with no key.

● LLM judging

299 real judgments on the top-300; 294 fact-validated reasoning lines. A local CLI backend + deterministic fallback — runs **with or without** any key.

● GCP — Plane C

Cloud Run for API + web, Secret Manager for the key, optional Vertex for Plane A. Documented and credential-gated — never needed to grade.

● The boundary

Every AI / network call lives in Plane A or C. The graded `rank.py` makes **zero** network, GPU, or LLM calls — proven by a no-socket test and a `--network none` run. NVIDIA and GCP are quality multipliers, never dependencies.

One command. Sealed. Twice → identical.

```
# the sealed graded reproduction docker run --rm --network none --cpus=4 --memory=16g \ indiaruns-sandbox ->
/out/submission.csv # PASS # or directly python rank.py --candidates candidates.jsonl --out submission.csv # 73.5
s · 2.01 GB peak RSS · CPU-only
```

- ◆ Run it twice and byte-diff — **identical** (PYTHONHASHSEED=0, OMP_NUM_THREADS=1, artifact bytes tracked by sha256 in the manifest).
- ◆ The vendored `validate_submission.py` runs **inside** `rank.py` — it refuses to write a non-conforming file.
- ◆ Nothing to trust on faith: **the grader reproduces the exact bytes.**

The hiring brain **Redrob** is already building toward.

● Monetization

Per-seat recruiter SaaS · usage-based ranking API for ATS & job boards
· premium co-pilot outreach & intelligence · enterprise on-prem for compliance-sensitive hiring.

● Virality

Shareable shortlists with auto OG images turn every result into a "make your own ranking" funnel. The honeypot-reveal drawer is inherently screenshot-worthy.

● Trust is the moat

Evidence-linked reasoning · honest AI-status pill · a sealed reproducible ranker · a structural anti-trap guarantee — exactly the trust hiring decisions require.

● Redrob fit

Plugs straight into the super-app — job matching across **15M+ live jobs**, hiring intelligence, the AI app store — on the path to **10–20M** monthly Indians.

TEAM & CONTACT

Team IndiaRuns

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The Code

Clean, complete, reproducible GitHub repository.

The Sandbox

One-click hosted Space that runs the real rank.py on screen.

The Results

Valid, honeypot-clean, fully reasoned top-100 from the real 100,000.

Build something real. Make hiring smarter.